Exercise 5.5

I. Construct the following parallelograms with the given measurements and find their area.

1. ARTS, AR=6*cm*, RT=5*cm* and $\angle ART = 70°$.
2. CAMP, CA=6*cm*, AP=8*cm* and CP=5.5*cm*.
3. EARN, ER=10*cm*, AN=7*cm* and $\angle EOA = 110°$ where $\overline{ER}$ and $\overline{AN}$ intersect at O.
4. GAIN, GA=7.5*cm*, GI=9*cm* and $\angle GAI = 100°$.

II. Construct the following rhombuses with the given measurements and also find their area.

(i) FACE, FA= 6 cm and FC = 8 cm

(ii) CAKE, CA=5 cm and ∠A = 65°

(iii) LUCK, LC = 7.8 cm and UK = 6 cm

(iv) PARK, PR = 9 cm and ∠P =70°

III. Construct the following rectangles with the given measurements and also find their area.

(i) HAND, HA = 7 cm and AN = 4 cm

(ii) LAND, LA = 8 cm and AD = 10 cm

IV. Construct the following squares with the given measurements and also find their area.

(i) EAST, EA = 6.5 cm

(ii) WEST, WS = 7.5 cm

SUMMARY

- Congruent figures are exactly the same in shape and size.
- Similar figures have the same shape but different sizes.
- In a right angled triangle, the square on the hypotenuse is equal to the sum of the squares on the other two sides. This is known as Pythagoras theorem.
- The three medians of any triangle are concurrent. The point of concurrence of the three medians in a triangle is called its Centroid, denoted by the letter G.
- The three altitudes of any triangle are concurrent. The point of concurrence of the three altitudes of a triangle is called as its Orthocentre, denoted by the letter H.
- The three perpendicular bisectors of the sides of any triangle are concurrent. The point of concurrence of the three perpendicular bisectors of a triangle is called as its Circumcentre, denoted by the letter S.
- The three angle bisectors of any triangle are concurrent. The point of concurrence of the three angle bisectors of a triangle is called as its Incentre, denoted by the letter I.
- A trapezium is a quadrilateral in which a pair of opposite sides are parallel.
- A parallelogram is a quadrilateral in which the opposite sides are parallel.
- Rhombus is a parallelogram in which all its sides are congruent.
- Rectangle is a parallelogram whose all its angles are right angles.
- Square is a parallelogram in which all its sides and angles are equal.

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